PlanArena: validated decision support and optimization for land-use change

Parcel-level impacts, likelihood and recommendations on official data for ten US cities

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Abstract

Planners decide land-use changes parcel by parcel. Planning support tools rarely show together what a change would do, whether zoning and the adopted plan allow it, how likely it is and which alternatives would do better, and they rarely test these outputs against what happened. PlanArena is an open decision support system that provides all of these for 4,995,208 parcels in ten large US cities, built from official land-use, plan and zoning data. Any edit is scored on 11 social, economic and environmental indicators in 45 ms (Manhattan) to 0.57 s (New York City). A redevelopment model estimates how likely the change is, an optimizer recommends plans, and voting rules compare how stakeholders would decide. Each component was tested on observed outcomes. In Manhattan, predicted effects of 2015–2025 changes correlated with recorded homes ($r = 0.840$) and emissions ($r = 0.857$), and redevelopment was ranked out of time with an AUC of 0.847–0.952. In seven cities, zoning and plans added little to predicting where development happens but helped predict what it becomes. Likelihood weighting brought projections of new homes from 76–193 to 0.81–2.64 times what was built. The optimizer found the exact optimum in 93% of small test areas.

Keywords: planning support systems; decision support; land-use optimization; urban digital twins; impact assessment; model validation

1 Introduction

Land-use change in US cities is decided parcel by parcel. Rezonings, amendments to comprehensive plans and approvals of conversions are justified by their expected effects on housing, jobs, services, the tax base and the environment. To weigh a proposed change, a planner needs to know five things about each affected parcel:

1. what exists;
2. what zoning permits;
3. what the adopted plan intends;
4. what is likely to happen;
5. what alternative would do better, and what each option would do across these aspects.

Planning support systems and urban digital twins aim to provide this information (Klosterman, 1999; Geertman and Stillwell, 2020; Batty, 2018; Lehtola et al., 2022).

Three gaps limit how far planners can rely on such tools.

- **Testing.** Their outputs are seldom tested against what happened. Reviews of planning support systems and urban digital twins find that claimed capabilities run ahead of demonstrated ones (Vonk et al., 2005; Lei et al., 2023; Azadi et al., 2025; Kruse et al., 2026). Scenario tools commonly project development at capacity, assuming that each site builds out to what the rules allow. In three Manhattan rezonings replayed in this paper, capacity projected 8,423 homes on development sites; 561 had been built by 2025.
- **Plausibility.** Land-use optimization generates plans that score well on chosen objectives (Malczewski, 2006; Rahman and Szabó, 2021; Zheng et al., 2023). It rarely asks whether the recommended changes are plausible, or how the choice among plans depends on the decision rule used to make it (Brandt et al., 2016; Feng et al., 2024; Lin et al., 2025).
- **Data.** Existing land use, zoning and the adopted plan are published by different agencies in different schemas. Some plans exist only as documents: Philadelphia's land-use plan exists only as maps in 17 PDF district plans.

This paper presents PlanArena, an open parcel-level decision support system, and tests each of its components on observed outcomes. The contributions are as follows.

1. **An integrated system.** For every parcel, PlanArena combines what exists, what is permitted, what is planned, what is likely and what is recommended. Any edit is scored on 11 indicators across three pillars (social, economic and environmental), with sensitivity ranges, at interactive speed. A multi-objective optimizer recommends plans within zoning and precedent, and the system compares how alternative decision rules would choose among them.
2. **Official data for ten cities.** The official existing land use, adopted plan and zoning for 4,995,208 parcels are put on one schema. A validated method turns a plan published only as vector PDF maps into a parcel layer.
3. **Validation of every component on observed data** (Table 3). This covers existing conditions, reproduction of environmental reviews, effects of observed change, forward prediction of redevelopment, a cross-city test of what zoning and plans add, scenario projections, and optimizer quality and plausibility.
4. **Design guidance from the evidence.** The results show which layers a tool should rely on for which question, and how it should present projections and decisions.

The scope is stated throughout. Impacts are validated against observed data in Manhattan; elsewhere the coefficients are national or regional. The stakeholder votes are simulated, not elicited. Usability with planners is not evaluated here.

2 Related work

Planning support systems and urban digital twins. Planning support systems combine land suitability, demand projection and allocation in scenario planning (Klosterman, 1999; Geertman and Stillwell, 2020), often within a geodesign workflow that links representation, change, impact and decision (Steinitz, 2012). Their limited use and uncertain usefulness are long-standing concerns (Vonk et al., 2005; te Brömmelstroet, 2013; Pelzer, 2017). Urban digital twins extend the idea to integrated, updatable city models (Batty, 2018, 2024; White et al., 2021; Lehtola et al., 2022; Ferré-Bigorra et al., 2022). Reviews report technical and

organisational barriers (Lei et al., 2023; Wu and Guan, 2025) and a gap between ambitions and demonstrated contributions to planning (Azadi et al., 2025; Luo et al., 2025; Kruse et al., 2026). PlanArena responds by attaching a test against observed data to each output it reports.

Land-use optimization and collective decisions. GIS multicriteria analysis and multi-objective optimization generate and rank land-use alternatives (Malczewski, 2006; Rahman and Szabó, 2021), recently with deep reinforcement learning (Zheng et al., 2023) and with explicit stakeholder interests (Feng et al., 2024; Lin et al., 2025). Planning problems have no single correct solution (Rittel and Webber, 1973), so the rule that turns stakeholder preferences into a choice matters. Bargaining (Nash, 1950) and voting rules have different properties (Schulze, 2011; Brandt et al., 2016). Bargaining has been combined with allocation in land reallocation (Hashemi et al., 2026). Simulated votes are not participation (Arnstein, 1969); PlanArena uses them only to show how sensitive a choice is to its rule.

Predicting development, and the role of regulations. Urban change models range from cellular automata (Clarke et al., 1997) and statistical site models (Landis and Zhang, 1998) to behavioural simulation (Waddell, 2002; Waddell et al., 2003) and agent-based models (Parker et al., 2003), with neighbourhood and temporal effects as priorities (Verburg et al., 2004; Weerasinghe et al., 2025). At the parcel level, models capture existing and historical neighbourhood conditions (Tepe and Guldmann, 2020), and machine learning complements econometric analysis of redevelopment (Carruthers and Wei, 2024). Redevelopment depends on land value relative to the existing structure (Rosenthal and Helsley, 1994; Dye and McMillen, 2007); in Chicago, smaller and older buildings with less lot coverage are replaced more often (Weber et al., 2006; Charles, 2013).

Regulation matters too, but it is not exogenous:

- rezoning follows development opportunity (Gabbe, 2018);
- causal studies of upzoning differ on its short-run effect on construction (Freemark, 2020; Dong, 2024; Greenaway-McGrevy and Phillips, 2023);
- plans are only partly implemented (Talen, 1996; Laurian et al., 2004; Brody and Highfield, 2005; Berke et al., 2006; Loh, 2011);
- recent land-change models integrate zoning and planning regulations (Hersperger et al., 2018; Domingo et al., 2021; Abolhasani et al., 2026), but do not test what those layers add to prediction on held-out history.

Validation. Land-change predictions should be compared with persistence at several resolutions (Pontius et al., 2008; Pontius and Millones, 2011), with transition probabilities validated directly (Black et al., 2023). Under spatial and temporal dependence, random cross-validation overstates skill (Roberts et al., 2017; Ploton et al., 2020), and models apply only where new data resemble their training data (Meyer and Pebesma, 2021). Uncertainty from many coefficients is better represented by varying them jointly (Saltelli et al., 2019).

3 The system

Figure 1 shows the architecture. A parcel twin holds each city's official layers. Four engines then answer the planner's questions:

- the impact engine: what would this change do?

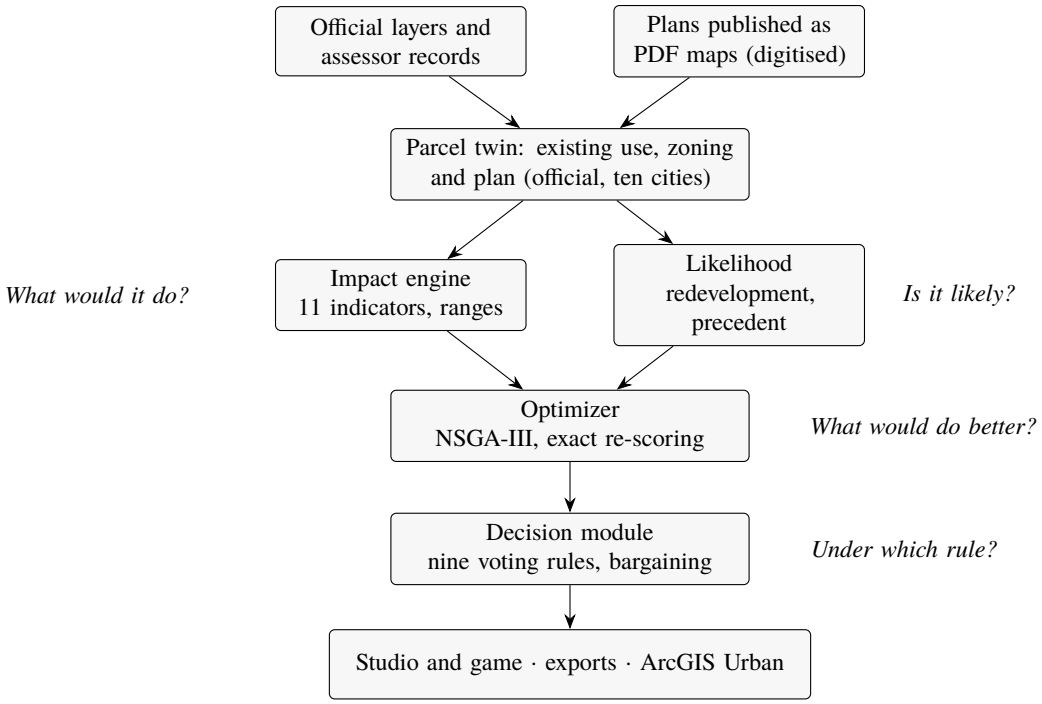


Figure 1: PlanArena's architecture. Every number the system shows comes from one of these components, and each component is tested on observed data (Table 3).

- the likelihood models: is it likely?
- the optimizer: what would do better?
- the decision module: who would choose it, under which rule?

The Studio, a browser application, lets a planner paint uses and floor-area ratios on parcels and see every indicator, flag and recommendation update (Figure 2).

3.1 Parcel twin and official layers

The twin covers the nine largest US cities other than San Jose, and Milwaukee (Table 1).

- **Existing land use** is the parcel land use published by the city or its regional planning agency, refined with assessor records where the layer groups uses.
- **The adopted plan** (future land use) and **zoning** are separate layers.
- **Classes.** Every source code is translated by an explicit crosswalk into 16 classes. These are three housing types, mixed use, retail, office, hotel, industry, warehousing, education, health, civic, open space, transportation, parking and vacant land, and they nest in nine analysis groups.
- **Buildings.** Floor area, homes, year built and assessed value come from assessor or city records, with estimated fields flagged. Jobs are fitted to Census LEHD block counts.

Every input is downloaded by script, with its URL and SHA-256 hash recorded. Philadelphia publishes its plan only as PDF maps. Appendix A describes how the maps are digitised: 15 of 17 plans were converted,



Figure 2: The Studio in Manhattan. Thirty parking lots and vacant parcels in East Harlem have been painted as housing (outlined on the map). The panel on the right shows the plan score, the social, economic and environmental pillars, and each indicator's change within a ten-minute walk with its sensitivity range. The palette on the left sets the use; the bar at the top switches between the plan being edited and the city's existing, planned and zoned use.

covering 83% of parcels, with median registration errors of 1.1–7.2 ft, and each was checked against a shifted-map control. Where a city publishes no plan as data, the plan layer is absent rather than estimated.

3.2 Impact engine

An edit sets a parcel's use, and optionally its floor-area ratio (FAR). The engine computes the 11 indicators of Table 2 for the parcels within a ten-minute walk and for the city.

- **Unchanged parcels** keep their observed values, so the baseline reproduces the records exactly.
- **Changed parcels** follow a hybrid delta rule. If only density changes and the record is consistent, the parcel's observed composition is scaled. If the use changes, the observed old use is removed and the modelled new use added. The rule was chosen on observed change (Section 5.2).
- **Derived quantities.** Homes follow from residential floor area and each parcel's floor area per home. Jobs follow from floor-area components and rates fitted to LEHD block counts. Emissions follow from energy use intensities and emission factors derived from New York's energy benchmarking data (Local Law 84) and its emission limits (Local Law 97). Runoff follows from surface coefficients. Assessed value changes by the difference between modelled values.
- **Access indicators** (parks, services and facilities within a walk, land-use mix, green cover) are recomputed on a 50 m raster.

Table 1: The ten cities: official parcel layers on one schema. Shares are of the city's parcels.

City	Parcels	Existing use (source)	Share	Adopted plan	Share	Zoning	Homes / Census
New York	856,634	MapPLUTO (City Planning)	100%	none citywide	–	100%	1.03
Los Angeles	672,444	SCAG 2020; county assessor	100%	General Plan (SCAG 2019)	100%	100%	0.99
Chicago	606,894	CMAP 2023; county assessor	100%	none citywide	–	96%	1.05
Houston	630,525	City (appraisal codes)	97%	none	–	–	0.95
Phoenix	510,871	County assessor; MAG	100%	General Plan	100%	99%	0.99
Philadelphia	547,410	City Planning Commission; OPA	100%	district plans (digitised PDF)	83%	100%	0.99
San Antonio	438,009	Bexar Appraisal District	100%	adopted area plans	61%	82%	0.99
San Diego	272,948	SANDAG 2025; county assessor	100%	General and community plans	100%	94%	1.02
Dallas	311,431	City 2026; DCAD	100%	forwardDallas! 2.0	99%	79%	0.98
Milwaukee	148,042	MPROP	100%	PDF only (not included)	–	99%	1.06
All	4,995,208						

Existing use is the city's or regional agency's parcel land use, refined by assessor records where the layer groups uses. Homes / Census: homes in the twin over Census 2020 housing units. Houston has no zoning. Milwaukee's area plans are published as PDFs that could not be downloaded by script.

Table 2: Indicators computed for every edit, and how each is validated (Manhattan).

Indicator	Pillar	Unit	Better	Validation
Homes	economic	units	higher	observed change
Jobs	economic	jobs	higher	existing (tract, block)
Tax base (assessed value)	economic	USD	higher	observed change
Park within 5-min walk	social	% of residents	higher	not validated
Shops & services within 5-min walk	social	adequacy 0-1	higher	not validated
Community facilities within 10-min walk	social	adequacy 0-1	higher	not validated
Land-use mix around homes	social	index 0-1	higher	not validated
Green & open space	environmental	acres	higher	direct accounting
Stormwater runoff	environmental	impervious-equiv. acres	lower	not validated
Building emissions	environmental	tCO2e / yr	lower	existing and observed change
Green cover near homes	environmental	% of land within 400 m	higher	not validated

Unchanged parcels keep their observed values; changed parcels follow the delta equations. Each indicator carries a sensitivity range from low and high coefficient variants. Access, mix, green cover and runoff are recomputed from the resulting state but have no observed after-change data to test against.

- **Ranges.** Every coefficient has a source and a status, and low and high variants give a sensitivity range for each indicator.

Indicators are combined into pillar scores with weights the planner can change. Scoring a 50-parcel edit takes 45 ms on Manhattan's 42,075 parcels and 0.57 s on New York City's 856,634, including the ranges. Impacts are computed for the nine analysis groups, for which the coefficients are calibrated and validated.

3.3 Likelihood, precedent and pattern

Three models tell the planner whether a change is plausible.

- **Redevelopment.** A gradient-boosted model (LightGBM) gives each lot's probability of redevelopment within one, three or five years, and its likely new use. It is trained only on outcomes known at the decision year. Its features describe the lot (use, size, built and permitted floor area, age, land share of value), its neighbourhood and its zoning.
- **Precedent.** A conversion is flagged when buildings of that use and size have not been converted to the proposed use in the past decade.

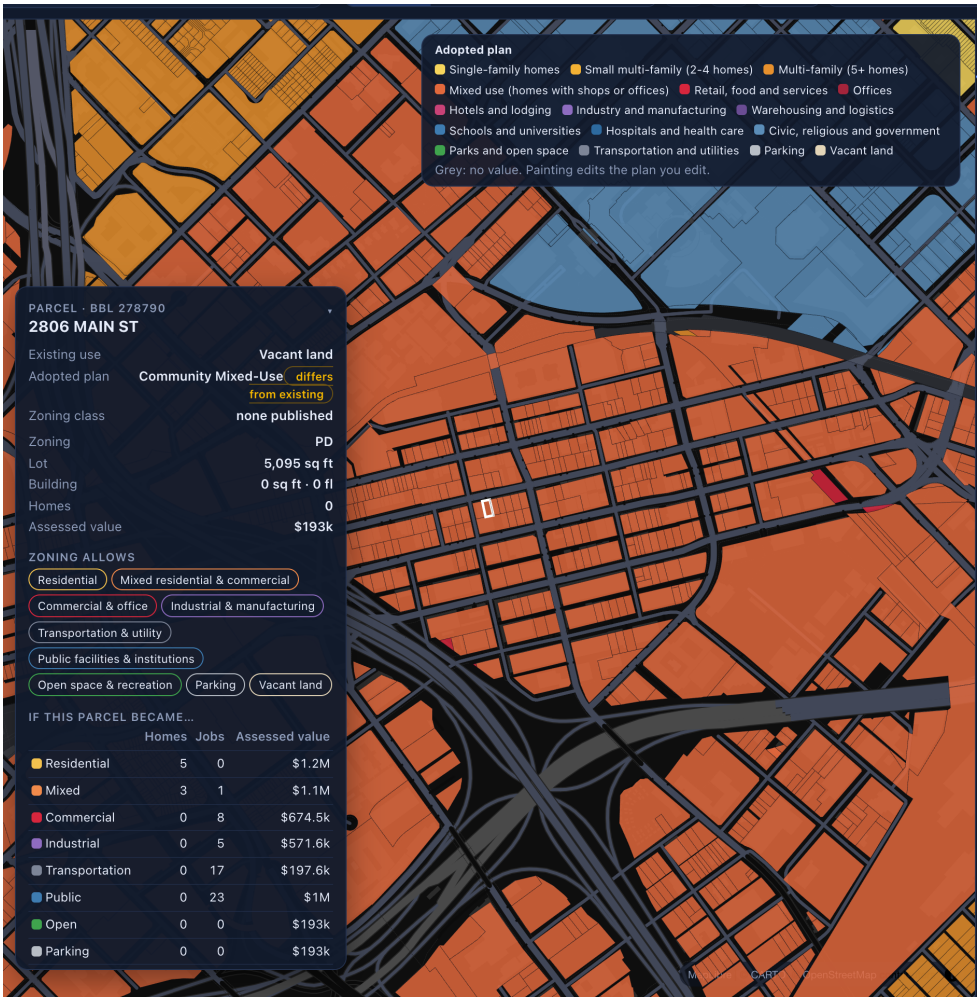


Figure 3: The parcel inspector on the adopted-plan layer (Dallas). For a vacant lot the plan designates as Community Mixed-Use, it lists the existing use, the plan designation (flagged as differing from the existing use), the zoning, the uses zoning allows and what each new use would yield in homes, jobs and assessed value.

- **Pattern consistency.** A graph neural network over parcel adjacency, combined with LightGBM, describes how well a use fits its surroundings. It describes existing patterns, not what is desirable.

Each edit is also flagged when it departs from the adopted plan or from what zoning allows (Figure 3).

3.4 Optimizer

The optimizer recommends plans that raise the weighted pillar score, minus a penalty per conversion, under a budget of changes.

- **Constraints.** Zoning is a hard constraint, and conversions without precedent are skipped. Optionally, moves are weighted by their likelihood.
- **Search.** Two methods are provided:
  - iterated best response over first-order estimates of each move, with exact re-scoring of the best moves;

Table 3: Validation design: each component of the system and its test against observed data.

Component	Question	Data	Section
Existing conditions	Does the twin reproduce what is there?	LL84 energy, LODES jobs, recorded homes	5.1
Planning calculations	Does it reproduce a review's projection from the review's own inputs?	FEIS site tables (two rezonings)	5.1
Effects of change	Given an observed change, does it predict the effect?	MapPLUTO 2015 vs 2025	5.2
Likelihood	Does it rank lots that redevelop, forward in time?	Rolling origins 2016–2025	5.3
Official layers	What do zoning and plans add, in other cities?	Seven city panels	5.4
Scenario projections	Are projected homes close to what was built?	Four cities; three rezonings	5.5
Optimizer	How close is the best plan found to the optimum?	Exhaustive search; 12 districts	5.6
Realism	Are recommended changes plausible?	2020 plans vs 2025 outcomes	5.6

  - a reference-point evolutionary algorithm (NSGA-III; Deb and Jain, 2014) that returns a Pareto set across the pillars.

Both report the best plan found, not a proven optimum.

3.5 Decision module

Planning choices are contested (Rittel and Webber, 1973). The decision module shows how the choice among candidate plans depends on the decision rule.

- **Stakeholder agents.** Five agents (residents, business, developers, environmental and city-finance interests) score each candidate from assumed utilities over the indicators.
- **Rules.** Nine voting rules, including Schulze's method (Schulze, 2011), which decides, are compared with the Nash bargaining solution (Nash, 1950) and with a residents' referendum by block.
- **Checks.** The module reports whether a Condorcet winner exists, whether one insincere ballot could change the outcome, and how robust the winner is to the assumed coefficients and seats.

The utilities are assumptions the planner can edit. The output is a simulated scenario, not public support (Arnstein, 1969). The optimizer's recommendation and the vote appear side by side in the Studio (Figure 4).

3.6 Interface and interoperability

Planners can work in the Studio in two ways: edit parcels by click, box, lasso or search; or upload plans as tables or GIS files, which are reviewed before they are applied. Studies are stored with a hash-chained audit log. Exports include a staff-report draft, CSV and GeoJSON files, and a provenance record of every coefficient. A phone version presents the same engine as a game. Any city or district can be exported to ArcGIS Urban: its space-use, zoning and land-use types, boundaries, and parcels with a PlanArena scenario.

4 Validation design

Each component is tested against observed data (Table 3). Wherever outcomes are predicted, models are trained on one period and tested on a later one. Uncertainty is estimated by resampling spatial tiles (Kunsch, 1989), and random splits are avoided (Roberts et al., 2017).

The cross-city test (Section 5.4) uses seven cities with parcel histories (Appendix Table 11).

- **Outcomes.** Development is defined as each source records it: a new building or a large increase in floor area (Los Angeles, Dallas, New York), a new building (Milwaukee), a change of use class between land-use inventories (Phoenix, San Diego), or a new-construction permit (Chicago).
- **Blocks.** Four feature blocks are added in turn: existing conditions (E), neighbourhood context within 400 m (C), zoning (Z) and the adopted plan (P).
- **Leakage controls.** Where a zoning or plan layer is current rather than historical but its changes are dated, a conservative variant drops lots whose zoning or plan dates from the window's start or later.

5 Results

5.1 Existing conditions and planning calculations

The twin reproduces Manhattan's existing conditions:

- **Emissions.** Modelled building emissions match 6,234 benchmarked buildings in a year not used for fitting (log–log $r = 0.807$, median error 31%, total 1.04 times metered).
- **Jobs.** Modelled jobs correlate with LEHD counts at tract level ($r = 0.843$) and less well at block level ($r = 0.727$).
- **Homes.** Given the observed use and floor area of 295 lots developed in 2015–2025, the estimated homes total is 1.04 times the recorded count ($r = 0.903$).
- **Environmental reviews.** Given the reviews' own sites, floor area and uses, PlanArena reproduces the projected homes of two rezoning reviews at 0.86 and 1.00 times the review figures.

5.2 Effects of observed change

The engine predicts the effects of real changes better than a simple baseline (Table 4). For 3,350 lots that changed in Manhattan between 2015 and 2025, each was given its 2015 state and its observed new use and floor area.

- **Correlation.** The predicted change in homes correlates with the recorded change at $r = 0.840$, against $r = 0.427$ for the mean change of the same transition type. Assessed value is at $r = 0.870$ against 0.148, and emissions of the 194 benchmarked changed buildings at $r = 0.857$ against 0.090.
- **Totals.** The engine overestimates the total change in homes (1.31 times observed). The baseline matches the total by construction.
- **Not predicted.** No method predicts job change at block level (engine $r = 0.068$).
- **Not validated.** Access, mix, green cover and runoff have no observed after-change data, and are marked as not validated in the Studio.

Table 4: Effects of observed change, Manhattan 2015–2025: given the 2015 state and the observed new use and floor area, the predicted change against the recorded change (Pearson r; homes: predicted over observed total).

Method	Changes	Homes r	Homes total	Value r	Emissions r
PlanArena	all	0.840	1.31×	0.870	0.857
	use change	0.872	1.29×	0.901	0.857
Mean change of the same transition	all	0.427	0.99×	0.148	0.090
	use change	0.435	0.97×	0.233	0.090

The baseline is cross-fitted: each change takes the mean observed change of other changes of the same type and size class. Emissions are tested on the changed buildings in the energy benchmarking data. Neither method predicts job change at block level.

5.3 Likelihood of redevelopment

The redevelopment model ranks lots forward in time.

- **Five years.** Trained on 2015–2020 outcomes and tested on 2020–2025, it reaches an AUC of 0.847, against 0.754 for the best simple baseline and 0.809 for a graph neural network.
- **Top of the ranking.** Of the top 1% of lots it ranks, 26% were redeveloped, against a base rate of 1.35%.
- **Calibration.** Its probabilities are well calibrated (expected calibration error 0.011).
- **Rolling origins.** Across 15 origins from 2016 to 2025, with horizons of one, three and five years, its out-of-time AUC is 0.847–0.952, above the best simple baseline at every origin.

5.4 What zoning and plans add, across seven cities

Existing conditions alone rank the lots that developed with an out-of-time AUC of 0.824–0.911 in six cities, and 0.571 in San Diego, where the one-year window recorded few changes (Appendix Table 12).

- **Context** adds the most (Table 5, Figure 5): from +0.020 to +0.116, significantly in five cities. Phoenix is the exception: its regional land-use series starts in 2022, so recent change is unknown in its training window.
- **Zoning** carries information of its own. Without context, it raises AUC significantly in five of seven cities. Developed lots were 2.1 to 9.5 times as likely as other lots to have a zoning class outside their existing use group.
- **Overlap.** Most of that information is shared with existing conditions and context: developed lots were also 3.5 to 19 times as likely to be vacant. Over context, zoning adds a significant gain in four of seven cities and the plan in three of four, at most +0.021 outside Phoenix. There, zoning adds +0.049 and the plan +0.054, and 45% of developed lots were vacant.
- **Conservative variant.** The plan gain in San Diego and the zoning gain in Chicago do not survive the conservative variant, so those gains came from lots rezoned or re-planned during the study.

The layers matter more for what a developing lot becomes (Table 6). In all six cities with a destination, the zoning class names the new class correctly more often than the existing use, by 12 to 64 percentage points. In Phoenix and San Diego development is a change of class by definition, so there the existing use is never right at class level. In Dallas the plan class is right for 80% of developed lots, against 61% for zoning.

Table 5: Gain in out-of-time AUC from adding a block, with 95% bootstrap intervals over 2 km tiles. Bold: the interval excludes zero.

City	C over E	Z over E+C	P over E+C	Z, conservative	P, conservative
Los Angeles	**+0.028 [0.020, 0.036]**	+0.002 [−0.001, 0.005]	0.000 [−0.004, 0.003]		
Dallas	**+0.020 [0.015, 0.025]**	**+0.004 [0.002, 0.005]**	**+0.002 [0.001, 0.003]**		
New York	**+0.023 [0.018, 0.029]**	**+0.004 [0.003, 0.007]**	–		
Milwaukee	−0.015 [−0.079, 0.048]	+0.029 [−0.031, 0.075]	–		
Phoenix	−0.066 [−0.139, 0.001]	**+0.049 [0.009, 0.091]**	**+0.054 [0.005, 0.100]**	**+0.055 [0.017, 0.088]**	**+0.041 [0.004, 0.073]**
San Diego	**+0.116 [0.053, 0.170]**	−0.003 [−0.022, 0.018]	**+0.021 [0.003, 0.039]**	−0.009 [−0.029, 0.015]	−0.011 [−0.033, 0.007]
Chicago	**+0.063 [0.049, 0.078]**	**+0.004 [0.001, 0.007]**	–	−0.001 [−0.005, 0.003]	–

Conservative: in each window, lots whose zoning or plan dates from the window's start or later are left out; possible where a current layer records those dates (Phoenix, San Diego, Chicago). Dallas's current layers carry no dates, so its zoning and plan gains are upper bounds.

Table 6: What developed lots became: share named correctly, as one of 16 classes / one of nine groups.

City	Developed	No change	Zoning class	Plan class	Model (blocks)
Los Angeles	1,883	57% / 79%	70% / 87%	70% / 87%	74% / 89% (E+C+Z+P)
Dallas	3,910	37% / 40%	61% / 65%	80% / 89%	86% / 95% (E+C+Z+P)
New York	4,002	24% / 37%	49% / 65%	–	54% / 70% (E+Z)
Milwaukee	78	18% / 18%	37% / 77%	–	63% / 76% (E+Z)
Phoenix	1,835	0% / 35%	64% / 74%	47% / 62%	75% / 81% (E+C+Z+P)
San Diego	202	0% / 52%	14% / 74%	27% / 70%	64% / 75% (E+C+Z+P)

In Phoenix and San Diego development is a change of class (to another group, or to more homes), so "no change" is never right at class level there. Models are cross-fitted over 2 km tiles among the developed lots. Chicago's permits do not record the new use.

A model fitted on other cities predicts less well than the city's own model in four of five cities (Appendix Table 13).

5.5 Scenario projections

Capacity projections overstate near-term housing by two orders of magnitude (Table 7).

- **Build-out.** In four cities, zoning and plan build-out exceed the homes added in the test window 76- to 193-fold. Part of the gap is definitional, since build-out is an end state rather than a two-year flow.
- **Likelihood weighting** brings the totals to 0.81–2.64 times the observed.
- **Trend.** A five-year trend ranks districts better: Spearman 0.51 to 0.81, against −0.37 to 0.55 for the likelihood-weighted projections, better in all four cities.

The Manhattan rezonings show the same contrast for single decisions (Table 8, Figure 6). Capacity projected 8,423 homes on development sites, or 5,207 pro-rated to 2025. The likelihood-weighted projection gave 585, against 561 built, and the observed count lies within its 95% range in two of three rezonings.

5.6 Optimizer and plausibility

Optimality. On 20 small study areas where every plan with up to two changes can be enumerated, first-order search alone found the exact optimum in 73% of areas, and with exact re-scoring in 93% (median 0.28 s). At district scale (Table 9), exact greedy search found the best plan in all 12 districts but took a median

Table 7: Homes added in the test window: projection over observed total (Spearman correlation across districts).

City	Observed	Zoning build-out	Plan build-out	Likely × zoning	Likely × plan	Five-year trend
Los Angeles	22,853	193× (0.10)	189× (0.24)	0.81× (0.46)	0.81× (0.52)	0.59× (0.70)
Dallas	10,926	141× (0.00)	127× (0.34)	2.64× (0.00)	1.69× (0.30)	1.25× (0.72)
New York	116,776	76× (0.34)	–	0.96× (0.55)	–	0.47× (0.81)
Milwaukee	3,798	115× (−0.23)	–	0.97× (−0.37)	–	0.32× (0.51)

Build-out: every lot at the homes per acre that the 90th percentile of lots of its zoning (or plan) class reaches in the city today. Likely: the build-out increment times the out-of-time probability of development (blocks E+C+Z+P where available). Trend: homes completed in the five years before the window, scaled to its length.

Table 8: Three Manhattan rezonings: homes on development sites by 2025, observed and projected.

Rezoning (adopted)	Sites	Built by 2025	Capacity	Pro-rated to 2025	Likelihood-weighted (95%)
East Harlem (2017)	127	288	4,117	3,294	234 (79–489)
Inwood (2018)	47	270	2,859	1,334	213 (0–652)
SoHo/NoHo (2021)	77	3	1,447	579	138 (8–304)
All three		561	8,423	5,207	585

Capacity is full build-out by the review's build year; pro-rated scales it linearly to 2025. The likelihood comes from a redevelopment model trained only on outcomes known by the adoption year.

25.87 s. First-order search with exact re-scoring reached a median 97.4% of that score in 1.14 s, the speed used in the Studio.

Plausibility. Recommended plans test plausibility against what happened next (Table 10).

- **Without realism layers**, 17% of recommended changes had no precedent in the city.
- **With the likelihood layer**, 21% of recommended parcels were redeveloped by 2025, against 12% without it, 1.4% of all lots and 5.2% of the likeliest fifth.
- **The cost** is a lower plan score: 2.73 against 8.27. What is plausible and what the objective prefers differ, and the Studio shows both.

5.7 Decision rules

In the simulated stakeholder scenario for 12 Manhattan districts, the decision rule changes the outcome.

- **Agreement.** Across the 12 districts, the nine rules never all agree; a Condorcet winner exists in seven.
- **Other choices.** The residents' referendum picks the council's plan in one district, and the council's winner is never the Nash bargaining solution.
- **Manipulability.** One insincere ballot could change the Schulze result in six districts.
- **Robustness.** The winner survives a median 76% of joint perturbations of coefficients and seats, and 54% of redrawn stakeholder weights.

These numbers describe the simulated scenario, not public opinion. They show that a recommendation should be reported with the rule that produced it, and with how it changes under others.

Table 9: Optimizer at district scale (12 Manhattan districts, up to ten changes): share of the best plan any method found.

Search	Median share of best	Districts best	Median seconds
surrogate top moves	92.6%	zero	0.02
first order	96.4%	one	0.13
first order plus exact polish	97.4%	one	1.14
greedy exact	100.0%	12	25.87
simulated annealing	94.8%	zero	7.54
random restarts 50	65.9%	zero	0.94

Table 10: Realism layers: plans recommended on the 2020 snapshot of every Manhattan district, and what happened by 2025.

Optimizer setting	Parcels	No precedent	Redeveloped by 2025	Plan score
no realism layer	529	16.6%	11.9%	7.52
precedent filter (default)	544	0.0%	12.1%	8.27
precedent filter + likelihood layer	118	0.0%	21.2%	2.73

For reference, 1.4% of all lots and 5.2% of the likeliest fifth were redeveloped. The plan score is the optimizer's own objective.

6 Discussion

What a planner gets. For any proposed change, PlanArena shows its effects across 11 indicators with ranges, and whether zoning and the plan allow it. It also shows how likely the change is, what the optimizer would recommend instead, and how the choice depends on the decision rule. Each of these outputs has a stated test against observed data (Table 3), and indicators without one are labelled. This answers, for one system, the gap between the claimed and demonstrated contributions of planning support systems and urban twins (Vonk et al., 2005; Azadi et al., 2025; Kruse et al., 2026).

Design guidance. The results support four practices for decision support tools.

1. **Use each layer for the question it answers.** Zoning and the adopted plan add little to predicting *where* development happens once existing conditions and context are known, but they help predict *what* it becomes, and they bound *how much* a site can hold. This is consistent with evidence that rezoning follows opportunity (Gabbe, 2018), that upzoning's short-run effect on construction varies (Freemark, 2020; Dong, 2024; Greenaway-McGrevy and Phillips, 2023), and that plans are partly implemented (Laurian et al., 2004; Berke et al., 2006).

2. **Report likelihood-weighted projections beside capacity,** and a trend baseline beside both. Capacity overstates near-term housing by orders of magnitude, and the trend ranked districts better than any projection.

3. **Constrain optimization by plausibility, and show the cost.** Realism layers change what is recommended and lower the objective; both views belong in front of the planner.

4. **Report decisions with their rule.** No rule agreed with all the others in any district of the simulated scenario.

Limitations.

- **Impacts** are validated only in Manhattan. In the other nine cities the engine uses national or regional coefficients, and its outputs there are unvalidated estimates.
- **Indicators.** Job change at block level, access, mix, green cover and runoff after a change have no observed test.
- **Cross-city test.** The windows are short (one or two years), outcome definitions differ by city, and in Dallas the plan and zoning are current layers without dates, so its gains are upper bounds.
- **Optimizer and votes.** The optimizer is heuristic, and the votes rest on assumed utilities.
- **Use.** Usability and use with planners are not evaluated here.
- **Causation.** Nothing here estimates what zoning or plans cause.

7 Conclusion

PlanArena gives planners the effects of any parcel-level land-use change across social, economic and environmental indicators, together with what is permitted, planned, likely and recommended. It runs on official data for ten large US cities at interactive speed, and each component is tested against what happened. The tests also refine how such systems should work:

- regulatory layers are best used for what development becomes and how much it could hold, not where it happens;
- likelihood-weighted projections are far closer to what was built than capacity, while trends rank districts better;
- recommendations change when they must be plausible, and with the rule used to choose among them.

The code, data manifest and results are open, so the system and its tests can be extended to other cities.

Data and code availability

The code, the result files behind every number, the figures and a manifest of every input (URL, retrieval date and SHA-256) are in the PlanArena repository; `make reproduce` rebuilds the data and results from public sources. Each city's ingestion report documents coverage, crosswalks and estimated fields. The Studio figures are regenerated by `planarena/tools/screenshots.py`. Owner and mailing fields in assessor records were never requested or read. No human participants were involved.

Declaration of AI assistance

[To be confirmed by the author.] The author used Claude (Anthropic) to write and test code, run analyses and draft text. The author designed the study, reviewed the code and results, read the cited sources and takes responsibility for the content. Every reference was checked against the record its DOI or ISBN resolves to.

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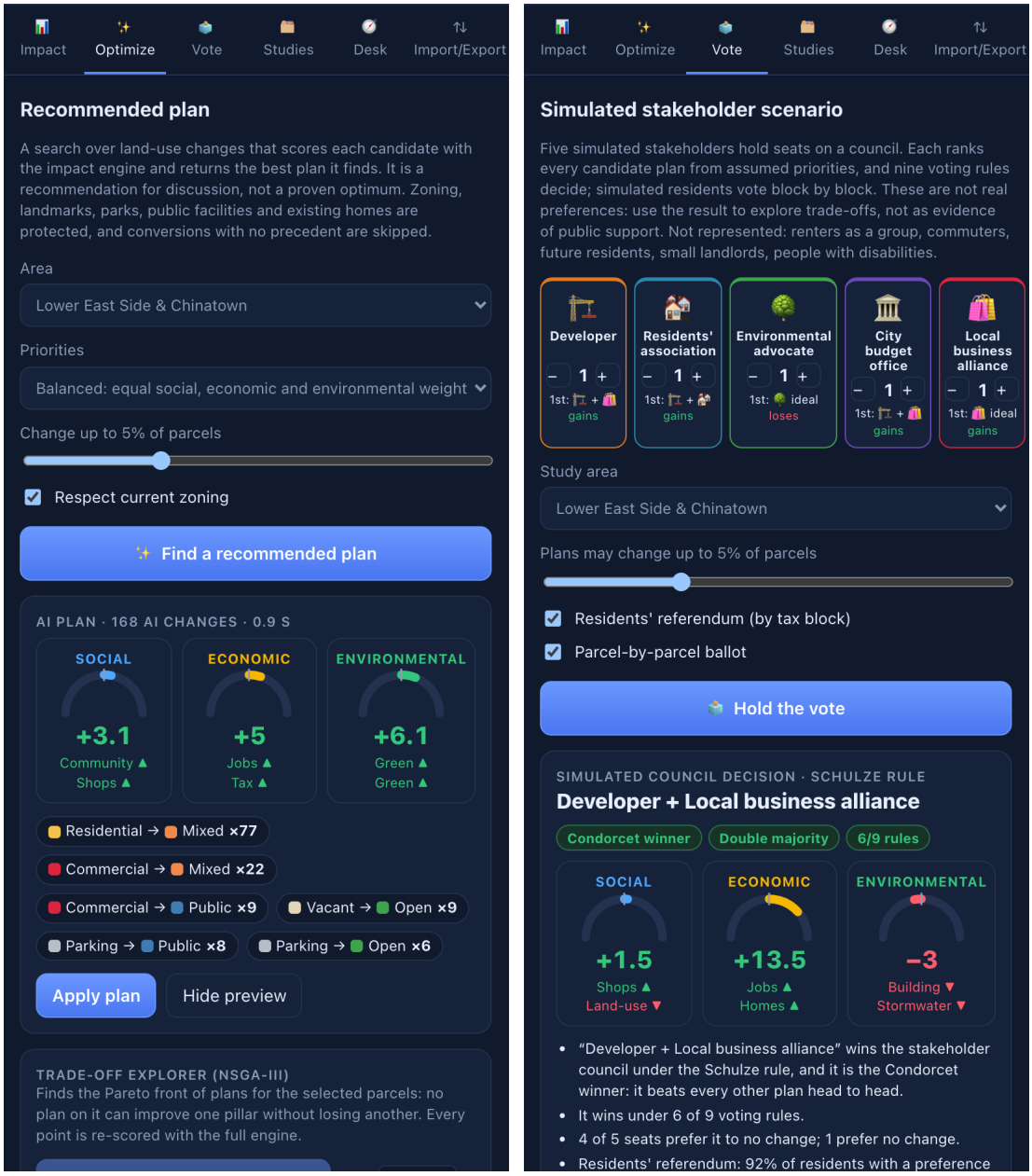


Figure 4: Optimizer and decision module for one Manhattan district. Left: the recommended plan, with its change in each pillar and the conversions it makes. Right: the simulated stakeholder council, the decision under Schulze's rule, whether the winner is a Condorcet winner and holds a double majority, and how many of the nine rules it wins.

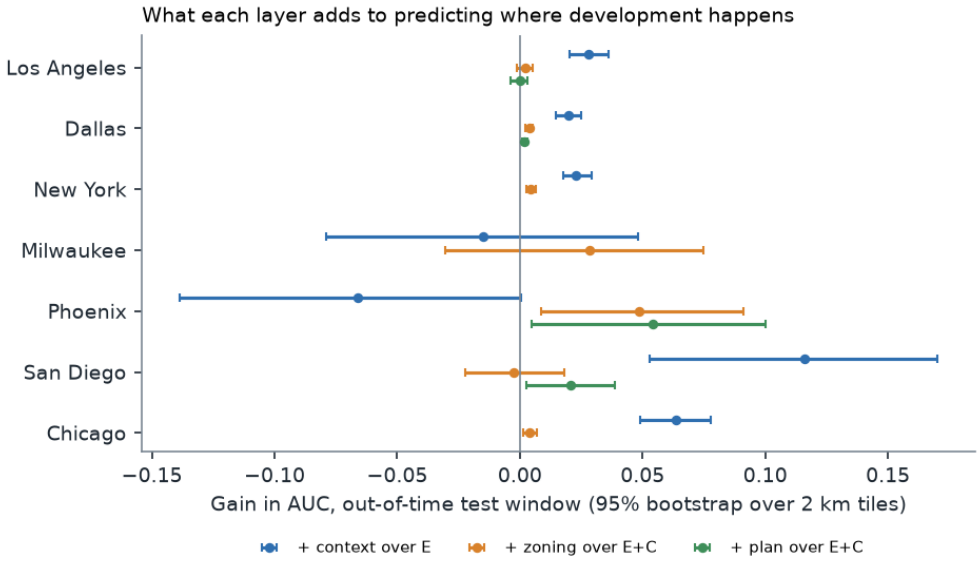


Figure 5: Gain in out-of-time AUC from adding each block, with 95% tile-bootstrap intervals.

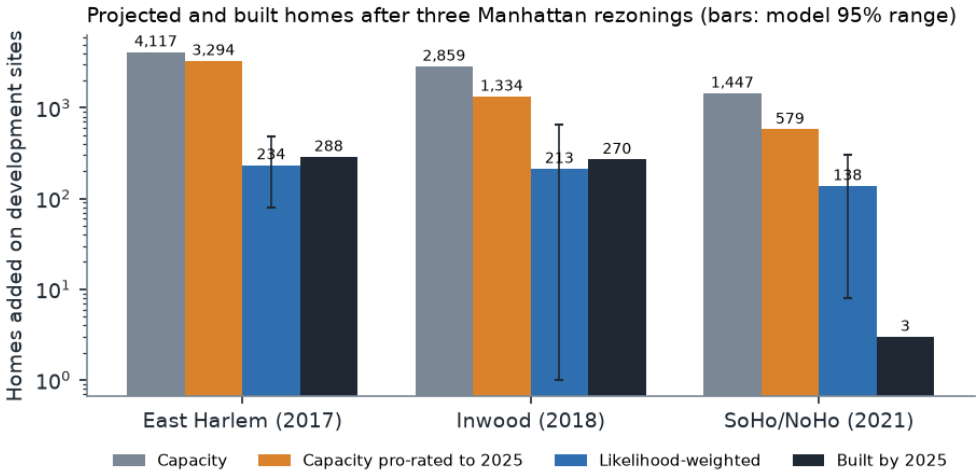


Figure 6: Homes on development sites after three Manhattan rezonings: capacity, capacity pro-rated to 2025, likelihood-weighted (with the model's 95% range) and built by 2025 (log scale).

A Digitising plans published as PDF maps

Philadelphia2035's land-use recommendations exist only as "Proposed Land Use" maps in 17 district plans covering 18 districts. The maps are vector drawings in which each lot is a filled shape. They are converted in six steps:

1. **Legend.** The legend's swatch colours and labels are read.
2. **Colours.** Map colours are matched one to one to legend colours by perceptual distance (CIELAB ΔE, Hungarian assignment), refusing pairs further apart than $\Delta E = 45$.
3. **Visibility.** Only shapes that the rendered page actually shows are kept (maps are often clipped to the district), and only the largest map frame on the page spread.
4. **Registration.** A similarity transform is fitted: the rendered footprint is first matched as an image to the district's parcels and outline (FFT translation, Dice score), then refined by iterative closest point (Reddy and Chatterji, 1996; Besl and McKay, 1992; Umeyama, 1991).
5. **Classification.** Each parcel takes the category of the smallest shape containing its interior point.
6. **Checks.** A map is used only if the median parcel-to-shape distance is at most 10 ft and its plan agrees with existing use clearly more often than the same map shifted 300 ft.

Maps from 15 of 17 plans passed, covering 16 districts and 83% of parcels.

- **Registration.** Median errors were 1.1–7.2 ft (median 3.2 ft).
- **Falsification test.** The digitised plans agree with the existing use on 60%–90% of parcels, against 42%–66% for shifted maps.
- **Failures.** The maps of South and West Park also draw neighbouring districts and failed registration.

Earlier map-processing work addresses scanned raster maps (Chiang et al., 2014; Luft and Schiewe, 2021; Milleville et al., 2022); vector exports allow the shapes to be read directly.

B Additional results

Table 11: Test panels. Models are fitted on the training window and scored on the test window.

City	Development	Source	Train	Test	Lots	Developed	Rate
Los Angeles	new building or floor area +25%	county assessor rolls	2021–2023	2023–2025	667,844	2,222	0.33%
Dallas	new building or floor area +25%	appraisal district files	2022–2024	2024–2026	291,841	3,916	1.34%
New York	new building or floor area +25%	MapPLUTO	2021–2023	2023–2025	856,901	4,064	0.47%
Milwaukee	new building	MPROP	2021–2023	2023–2025	147,539	85	0.06%
Phoenix	change of use group	MAG land-use inventory	2022–2024	2024–2026	503,635	1,835	0.36%
San Diego	change of use group	SANDAG land-use inventory	2023–2024	2024–2025	272,281	202	0.07%
Chicago	new-construction permit	building permits	2021–2023	2023–2025	606,294	1,548	0.26%

Lots already under construction at the start of a window are left out. Floor area +25% also requires +2,000 sq ft. Change of use group: the lot changes to a built use of another of the nine groups, or gains homes.

Table 12: Where development happens: out-of-time AUC by block set. E existing conditions, C neighbourhood context, Z zoning, P adopted plan; – where the city has no such layer.

City	E	E+C	E+C+Z	E+C+P	E+C+Z+P	Best set (any)
Los Angeles	0.864	0.892	0.894	0.892	0.894	E+C+Z+P
Dallas	0.898	0.918	0.922	0.920	0.923	E+C+Z+P
New York	0.892	0.914	0.919	–	–	E+C+Z
Milwaukee	0.911	0.896	0.924	–	–	E+Z
Phoenix	0.859	0.793	0.842	0.847	0.853	E+Z+P
San Diego	0.571	0.687	0.684	0.707	0.703	E+C+P
Chicago	0.824	0.888	0.891	–	–	E+C+Z

Best set: the highest AUC among all fitted combinations, including those without context.

Table 13: Transfer: AUC of a model fitted on the other cities' training windows / the city's own model, on the city's test window.

City	E	E+C	E+C+Z	E+C+Z+P
Los Angeles	0.719 / 0.864	0.817 / 0.892	0.828 / 0.894	0.837 / 0.894
Dallas	0.844 / 0.898	0.877 / 0.918	0.878 / 0.922	0.877 / 0.923
New York	0.825 / 0.892	0.874 / 0.914	0.868 / 0.919	0.866 / 0.919
Milwaukee	0.959 / 0.911	0.965 / 0.896	0.966 / 0.924	0.967 / 0.924
Chicago	0.742 / 0.824	0.808 / 0.888	0.804 / 0.891	0.822 / 0.891

Cities whose outcome is a building or a permit. A missing plan layer is coded as unknown.

Table 14: Out-of-time average precision and precision among the top 1% of lots.

City	Base rate	AP, E	AP, all	All blocks	Top 1%, E	Top 1%, all
Los Angeles	0.33%	0.139	0.147	E+C+Z+P	11.2%	12.8%
Dallas	1.34%	0.115	0.237	E+C+Z+P	12.6%	29.6%
New York	0.47%	0.073	0.122	E+C+Z	11.1%	15.1%
Milwaukee	0.06%	0.024	0.031	E+C+Z	2.3%	2.8%
Phoenix	0.36%	0.091	0.054	E+C+Z+P	9.7%	8.4%
San Diego	0.07%	0.002	0.002	E+C+Z+P	0.7%	0.6%
Chicago	0.26%	0.013	0.043	E+C+Z	2.4%	6.1%